

Anti-arrhythmic Drugs

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Pr.201

CARDIAC ARRHYTHMIAS

DEFINITION:

- ❖ *Deviation of either HR or rhythm from normal.*

SYMPTOMS:

- ❖ *Palpitation*
- ❖ *Dizziness*
- ❖ *Syncope*

DIAGNOSIS:

- ❖ *serial standard EKG,*
- ❖ *EKG during and after exercise*
- ❖ *Psychological stress, Holter's monitoring*

Cardiac arrhythmia: pathogenesis

- Myocardial infarction (MI): hypoxia
- Digitalis toxicity
- Catecholamines
- Anesthetics
- Hyperthyroidism
- Electrolyte imbalances: hypokalemia

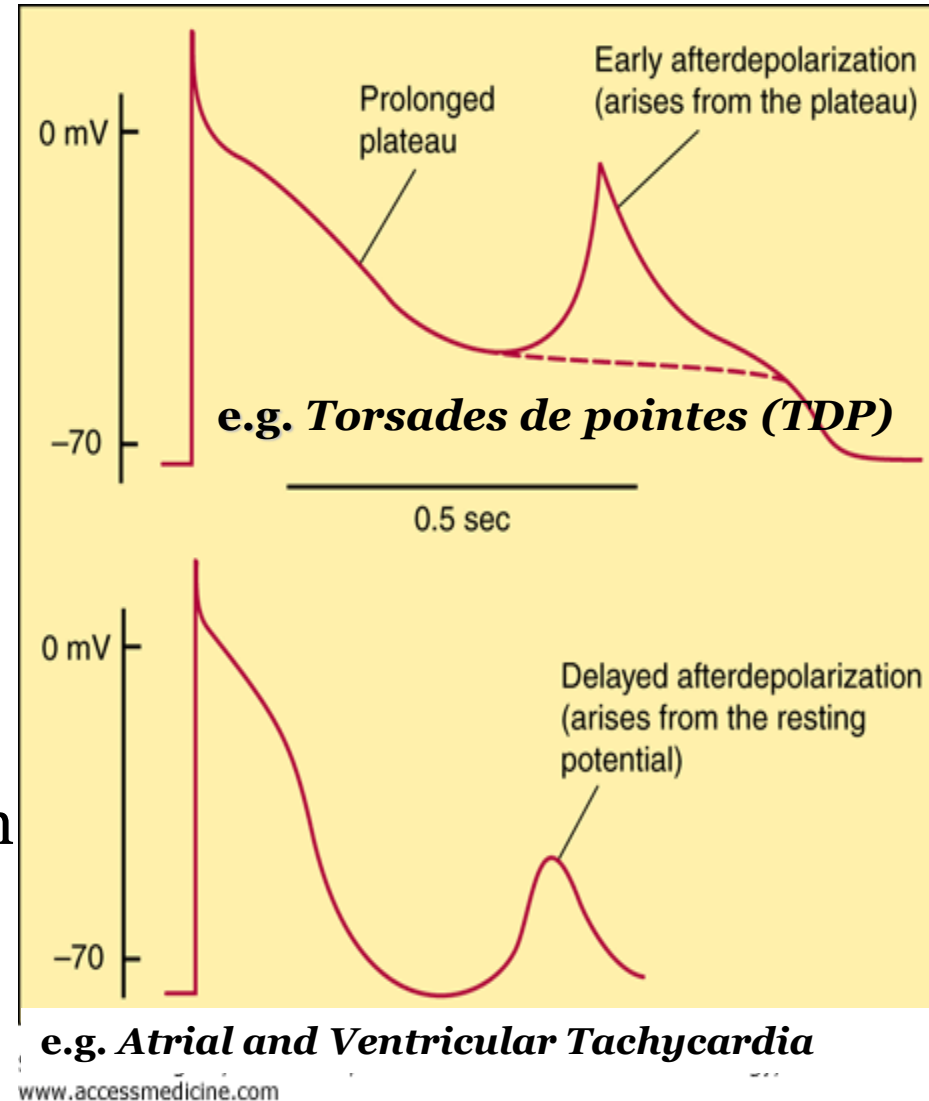
MECHANISM OF ARRHYTHMIAS

I. Abnormal impulse formation (spontaneous depolarization)

I.1 Enhanced automaticity of normally automatic cells (SA node)

I.2 Generation of impulses in normally non-automatic cells

- ❖ ‘triggered activity’ due to afterdepolarizations
 - early afterdepolarization (EAD)
 - delayed afterdepolarization (DAD)



MECHANISM OF ARRHYTHMIAS

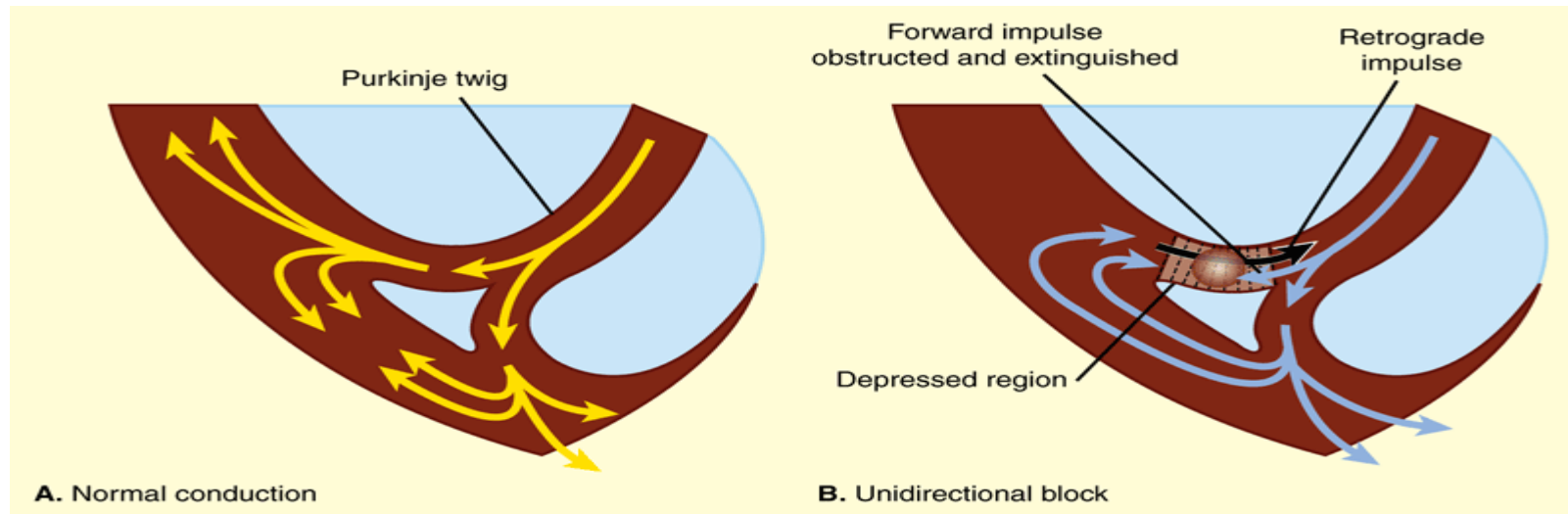
II. ALTERED CONDUCTION (Reentry mechanism)

II.1 Slowing and block

II.2 Re-entry

II.2.1 random re-entry (*functional*)

II.2.2 ordered re-entry (*anatomical*)



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology*, 12th edition: www.accessmedicine.com

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Classifying Antiarrhythmic Drugs

Aim of therapy is to reduce ectopic pacemaker activity and modify conduction or refractoriness in re-entry circuits to disable circus movement

Class 0	HCN Channel Blockers
Class I	Voltage-gated Na ⁺ Channel Blockers
Class II	Autonomic Inhibitors/Activators
Class III	K ⁺ Channel Blockers
Class IV	Calcium Channel Blockers

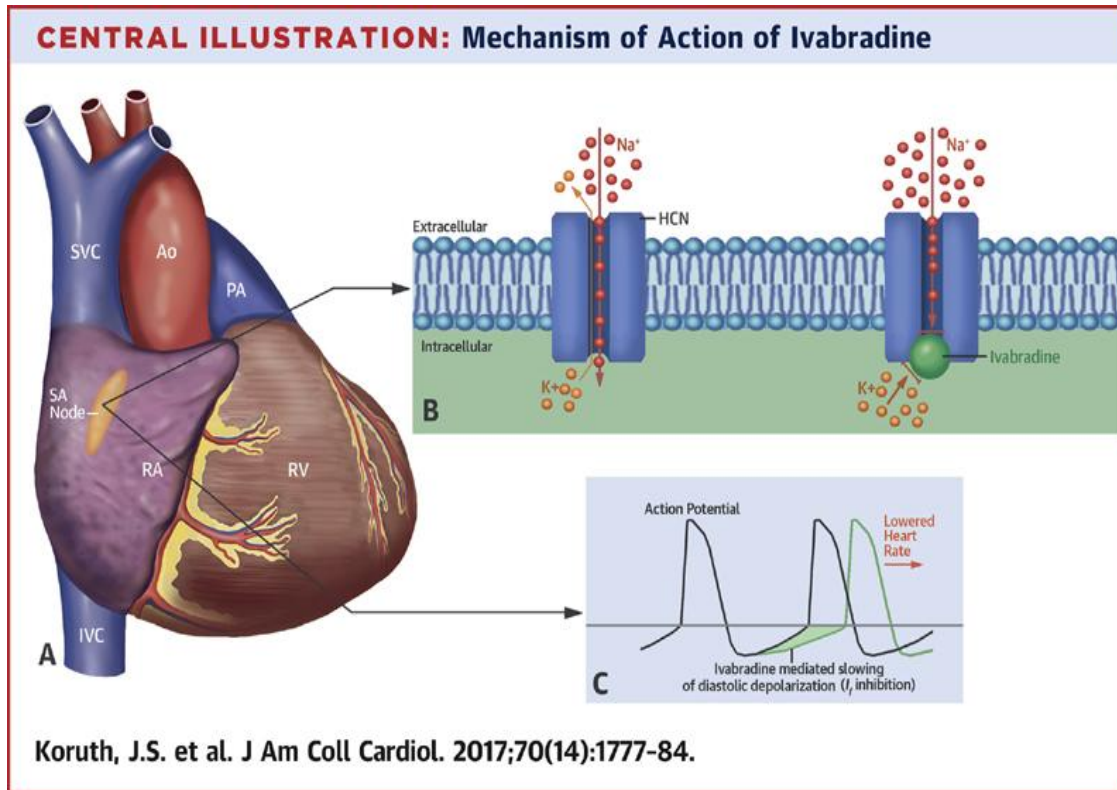
Table 1 Singh–Vaughan Williams classification of antiarrhythmic drugs and their clinical use. ACE, angiotensin-converting enzyme; AVNRT, atrioventricular node re-entry tachycardia; MI, myocardial infarction; PVC, premature ventricular contraction; SVT, supraventricular tachycardia.

Antiarrhythmic class	Examples	Clinical use
Class 0—HCN channel modulators	Ivabradine	Heart rate control for tachycardia
Class 1—voltage-gated Na ⁺ blockers		
1a—intermediate dissociation kinetics	Quinidine, procainamide	SVT, AF, VT/VF; not commonly used
1b—fast dissociation kinetics	Lidocaine, mexiletine, phenytoin	VT, particularly after MI
1c—slow dissociation kinetics	Flecainide	AF and atrial flutter
1d—late Na ⁺ channel (I _{NaL}) blocker	Ranolazine	AF treatment and prevention
Class 2—autonomic modulators		
β-adrenergic receptor blockers	Non-selective: propranolol Selective: metoprolol, esmolol	Prevention of supraventricular tachyarrhythmia Rate control of atrial and ventricular tachyarrhythmias
Muscarinic (M ₁) receptor blockers	Atropine, glycopyrrolate	Bradycardia, atrioventricular block
Muscarinic (M ₂) receptor activators	Digoxin	Rate control of AF
Adenosine – A ₁ receptor activators	Adenosine	SVT
Class 3—K ⁺ channel blockers		
Selective I _{Kr} blockers	Sotalol, ibutilide	Supraventricular arrhythmias, including AF, AVNRT, PVC, VF, VT; rate control of AF
Non-selective (I _{Kr} and I _{Ks}) blockers	Amiodarone	Termination of AF
Multi-ion channel blocker	Vernakalant	Paroxysmal SVT, rate control of AF and atrial flutter
Class 4—L-type Ca ²⁺ channel blockers	Verapamil, diltiazem	
Class 5—mechanosensitive channel blockers	No drugs available clinically	
Class 6—gap junction channel blockers	No drugs available clinically	
Class 7—upstream target modulators	ACE inhibitors and angiotensin II receptor blocker	

Class O : HCN Channel Blockers

Ivabradine

Mechanism of Action



- Blocks the channel responsible for the cardiac pacemaker current, $I(f)$, which regulates heart rate.
- This results in prolonged diastolic time and reduced heart rate.
- Used to reduce HR in **inappropriate sinus tachycardia** or when **sinus tachycardia accompanies cardiac failure**.

Class O : HCN Channel Blockers

Ivabradine

Adverse Effects

- **Most common:** bradycardia, atrial fibrillation, high blood pressure, and phosphenes (visual disturbances)

Contraindications

- Decompensated heart failure
- Blood pressure less than 90/50 mm Hg
- Conduction abnormalities (eg, sick sinus syndrome, sinoatrial block, or third-degree AV block)
- Severe liver impairment
- Patients taking cytochrome P450 3A4 (CYP3A4) inhibitors
- Resting heart rate less than 60 bpm before therapy is initiated

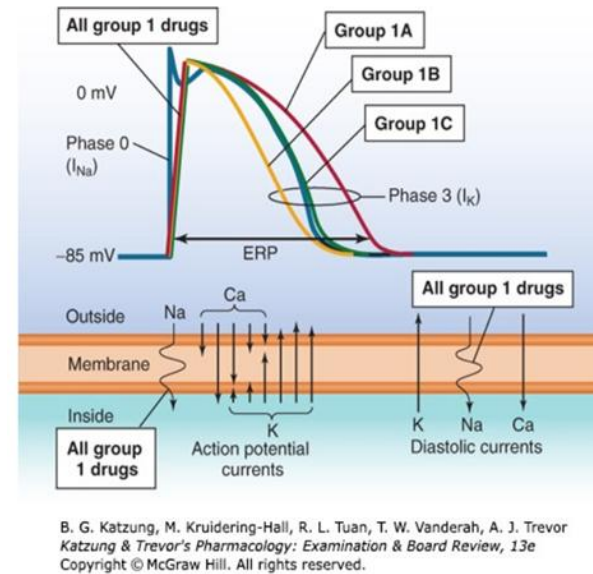
Class I – Na⁺ channel blockers

Class	Na ⁺ -Channel blockade	APD shorten	Repo. time
I-A	+2	+1	prolonged
I-B	+1	+3	shortened
I-C	+3	NC	NC

APD ≡ *Action Potential Duration*
Repo. Time ≡ *Repolarization Time*

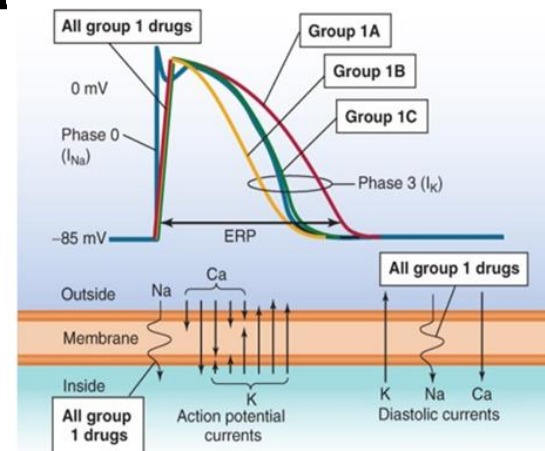
Class IA-quinidine, procainamide disopyramide

- ♥ Reduced $V_{\max}$ of cardiac AP- decrease conduction may abort re-entry **
- ♥ Depressed phase 4 depolarization in SA-node & ectopic pacemaker cells -inhibit triggered activity arising from DADs or EADs
- ♥ Prolonged repolarization- enhanced re-entry**
- ♥ Prolonged APD in Atrium, Ventricle and His Purkinje fibers



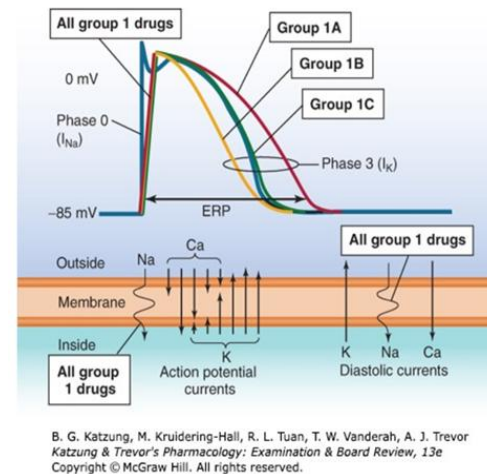
Class IB-lidocaine, mexiletine, phenytoin/tocainide

- Rapid interaction with inactivated state Na^+ channels
- Prolonged phase 0 depolarization
- Prolonged phase 4 depolarization in SA node
- Shorten APD



Class IC-flecainide/propafenone/moricizine

- Slow interaction with activated state of Na^+ channel blockers
- No effect on APD and QT interval
- Prolonged phase 0 depolarization ($\downarrow$ AV conduction)
- Prolonged phase 4 depolarization in SA node
- Propafenone has a weak β -blocking effect.



Therapeutic use: class IA, IB and IC Na^+ channel blockers

Used in all types of arrhythmias

Procainamide: 2nd or 3rd choice of drugs (lidocaine & amiodarone) for ventricular arrhythmias with acute MI

Lidocaine: 1st choice of ventricular tachycardia (VT) with acute MI & ventricular fibrillation (VF)

Amiodarone: use widely in treatment of arrhythmias; atrial fibrillation (blocked Na^+ , Ca^{2+} , K^+ channels & β -receptors)

Adverse drug reaction: class I Na^+ channel blockers

Drug-induced arrhythmia: especially quinidine (common)

↑APD (QT prolongation): can cause torsades de pointes

Hyperkalemia: exacerbates cardiotoxicity of all class I

Overdoses and treatment

Sodium lactate (reverse drug-induced arrhythmias)

Sympathomimetics (reverse drug-induced hypotension)

Class ID - Ranolazine

- Approved for treatment of chronic angina. However, this drug has potent antiarrhythmic activity.
- Inhibits I_{kr} , I_{ks} , late I_{Na} , I_{Ca} , and I_{Na-Ca}
- It appears to be useful for treatment of both **atrial fibrillation and ischemia-induced ventricular arrhythmias**.
- **ADR:** constipation, headache, dizziness, nausea, bradycardia, orthostatic hypotension, palpitations & QT prolongation

Class IIa β -Adrenoceptor Blockers

❖ $\downarrow$ HR

❖ $\downarrow$ AV conduction (prolonged phase 4 depolarization:

*** *treat re-entrant arrhythmias in AV-node*

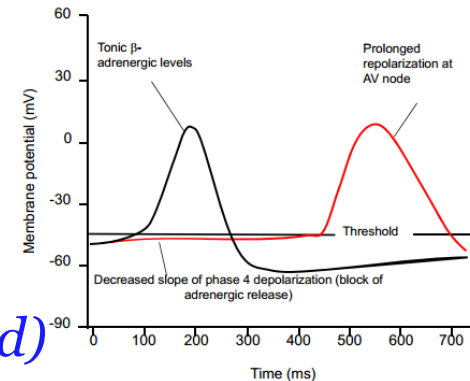
*** *control ventricular rate in Atrial flutter & Atrial Fibrillation*

❖ $\downarrow$ Na^+ & Ca^{2+} current:

*** *inhibits DADs and EADs*

$\downarrow$ Mortality rate after MI (*stress related*)

❖ Treat Na^+ -channel blockers-induced arrhythmias



Class IIa β -Adrenoceptor Blockers

- Metoprolol, bisoprolol, esmolol, atenolol, landiolol, nebivolol, etc.
- Sotalol has K⁺ channel blocking actions
- **Drug of choice: Control ventricular rate** in atrial fibrillation and atrial flutter
- Drug of second choice: paroxysmal supraventricular tachycardia (PSVT)

Side effect:

- β -adrenergic blockers: bronchospasm, hypotension, cold extremities, fatigue and impotence
- Negative inotropic effects
- CNS: insomnia and depression

Class IId M2 receptor activators

Digitalis glycosides

- Vagotonic action
- Hyperpolarization
- Shortening of atrial action potentials
- Increases in AV nodal refractoriness
- Used for **control ventricular rate in atrial fibrillation & atrial flutter** (second choice from β -blockers & Ca^{2+} channel blocker)

Side effects

- Proarrhythmia
- Electrolyte abnormalities
- Nausea/vomiting, lethargy, visual changes

Class IIe (Adenosine A1 receptor activator [adenosine])

- ❖ K⁺-channel Opener in atrium, SA & AV node
- ❖ Hyperpolarized membrane, indirectly inhibit Ca²⁺-channels
- ❖ Decreased AV conduction
- ❖ Increased PR intervals
- ❖ First line agent for terminating narrow complex PSVT due AV-reentrant tachycardia
- ❖ Contraindicated in asthmatic patients 2nd, 3rd degree AV-block

Class IIe (Adenosine A₁ receptor activator [adenosine])

- Very rapid onset of action
- Short half-life (seconds)

Side effects

- Atrial fibrillation/ sinus arrest
- Bronchospasm
- Flushing/headache
- Hypotension
- Nausea

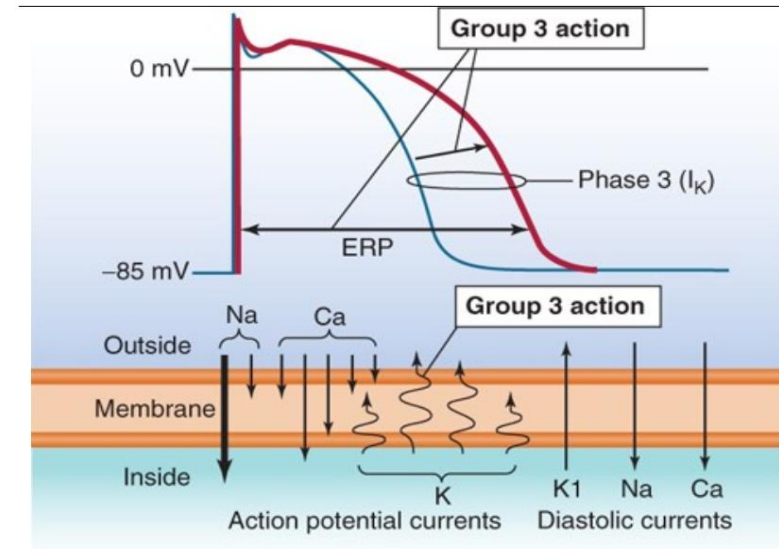
III. Potassium Channel Blockers

Mechanism of actions:

- ♥ *Decrease automaticity*
- ♥ *Delay depolarization - ↑ APD*
- ♥ *Increase refractoriness*

Other actions:

- ♥ Sotalol
 - ♥ *Also blocks β -adrenoceptor*
- ♥ Amiodarone, dronedarone
 - ♥ *also block Na^+ & Ca^{2+} channels and β -blocker*
- ♥ Ibutilide & Dofetilide
 - ♥ *pure class III agents*



B. G. Katzung, M. Kruidering-Hall, R. L. Tuan, T. W. Vanderah, A. J. Trevor
Katzung & Trevor's Pharmacology: Examination & Board Review, 13e
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Therapeutic uses: class III K⁺ channel blockers

- **Amiodarone (oral & IV):** VT, highly effective in supraventricular arrhythmias (SVT), atrial fibrillation
- **Sotalol:** life-threatening ventricular arrhythmias & maintenance of sinus rhythm in atrial fibrillation

ADR

- Torsades de pointes (TDP)
- **Amiodarone:** hypotension, QT prolongation, Torsade de Pointes, pulmonary fibrosis, myocardial depress (bradycardia and heart block), hepatic dysfunction, corneal microdeposits, photosensitization, hyperthyroidism or hypothyroidism

IV- Calcium Channel Blocker

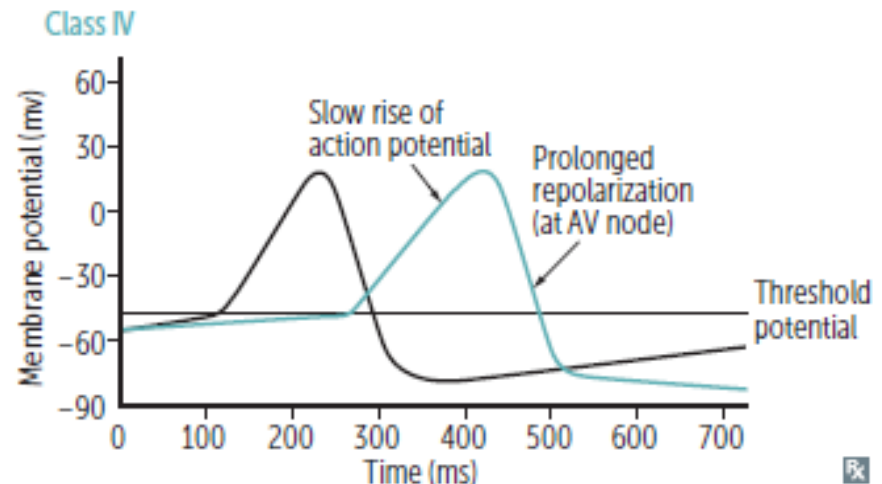
- ❖ Block L-typed voltage-dependent Ca^{2+} channels in SA & AV node
- ❖ Only **verapamil, diltiazem and etripamil (intranasal)** block Ca^{2+} channels in cardiac cells at clinically used doses.
- ❖ Delayed phase 4 & phase 0 depolarization in SA & AV nodes
automaticity & AV conduction, HR, ERP, PR interval
- ❖ Block VT due to DADs (*rare form*)

- ❖ Major side effects:
Verapamil, Diltiazem - hypotension (*with bolus injection*)

IV- Calcium Channel Blocker

Therapeutic use

- Effective only in atrial arrhythmias
- Treat re-entrant arrhythmias in AV-node & reduce ventricular rate in atrial flutter and atrial fibrillation.
- Second drugs of choice for the treatment of PSVT
- Not effective in VT.



Magnesium sulfate

- Mechanism of action is unknown
- Ca^{2+} channels blocker is possible
- **Used in:**
 - Digitalis -induced arrhythmias
 - Torsade de pointes
 - Ventricular tachycardia from hypomagnesemia

Principle in the Clinical Use of Antiarrhythmic Drugs

■ Identify and Remove Precipitating Factors

- *Hypoxia*
- *Hypo K⁺*
- *Myocardial Infarction*
- *Drugs: Digitalis, Theophylline, Erythromycin etc*

■ Established the Goals of Treatment

- Some arrhythmias should not be treated
- Symptoms due to arrhythmias
- Choosing among therapeutic approaches

■ Minimized Risk

- Antiarrhythmic can cause arrhythmias
- Monitoring plasma concentration
- Patient-specific contraindication

Questions ?

